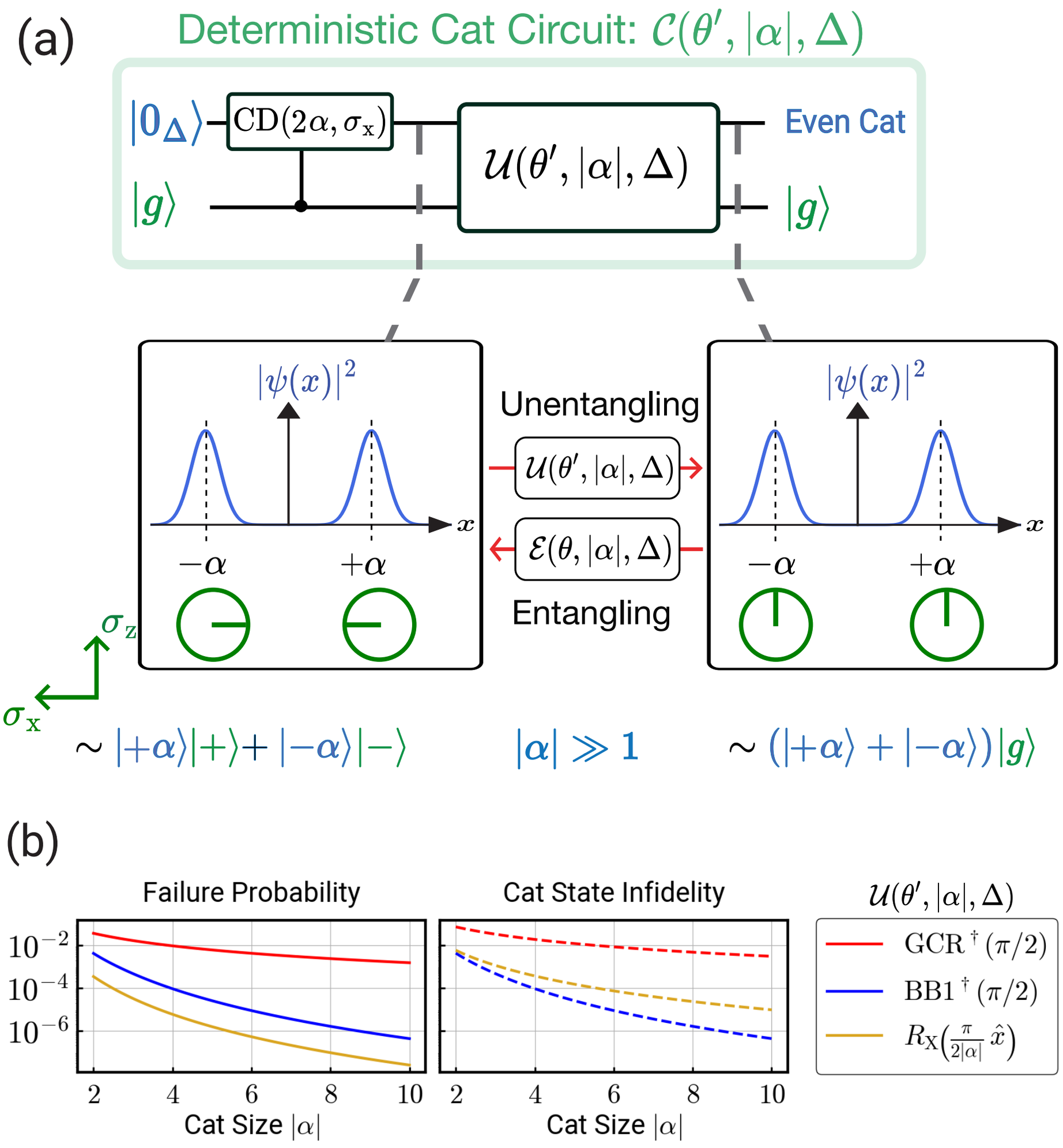

(a)
Deterministic Cat Circuit: $\mathcal{C}(\theta', |\alpha|, \Delta)$
$|0_\Delta\rangle$
$\mathrm{CD}(2\alpha, \sigma_x)$
$|g\rangle$
$\mathcal{U}(\theta', |\alpha|, \Delta)$
Even Cat
$|g\rangle$
$|\psi(x)|^2$
x
$-\alpha$
$+\alpha$
Unentangling
$\mathcal{U}(\theta', |\alpha|, \Delta)$
$\mathcal{E}(\theta, |\alpha|, \Delta)$
Entangling
$|\psi(x)|^2$
x
$-\alpha$
$+\alpha$
σ_z
σ_x
$\sim |+\alpha\rangle|+\rangle + |-\alpha\rangle|-\rangle$
$|\alpha| \gg 1$
$\sim (|+\alpha\rangle + |-\alpha\rangle)|g\rangle$
(b)
Failure Probability
Cat State Infidelity
10^{-2}
10^{-4}
10^{-6}
2
4
6
8
10
Cat Size $|\alpha|$
2
4
6
8
10
Cat Size $|\alpha|$
$\mathcal{U}(\theta', |\alpha|, \Delta)$
GCR † $(\pi/2)$
BB1 † $(\pi/2)$
$R_X\left(\frac{\pi}{2|\alpha|}\hat{x}\right)$